Numerical Simulation of 3D Printing of Infrastructure Materials

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ABSTRACT

3D printing is poised to become a disruptive force for the terrestrial construction industry, and it is also the most likely technology that will make possible the safe and reliable development of extraterrestrial infrastructure. While more and more researchers are working on developing novel and appropriate mixes, as well as 3D printing technologies, only a few and somewhat limited computational models are available in the literature for the design and analysis of 3D printed structures. Nevertheless, engineers need simulation tools for the design and optimization of any structure, including 3D printed. The main objective of this paper is to highlight the formulation of the discrete fresh concrete model (DFCM). DFCM is a particle-based fresh cementitious material model that simulates the rheological and mechanical behavior of material accurately before, during, and after printing to optimize the printing process and the material flow. For DFCM, the interaction among spherical particles is governed by visco-plastic constitutive equations, which, unlike conventional particle models that use force-displacement relationships, are based on stress-strain relationships. This allows a mathematical formulation enabling a seamless transition from the description of fresh concrete to the one of hardened concrete.

INTRODUCTION

The significant influence of fresh concrete characteristics (flowability and workability) on its mechanical responses and on its printability, have inspired the researchers to develop a variety of simulation techniques for standard concrete at its earliest ages (Christensen, 1991; Mori and Tanigawa, 1992; Kurokawa et al., 1996; Thrane et al., 2004; Roussel, 2006; Roussel et al., 2007; Tregger et al., 2008; Gram and Silfwerbrand, 2010; Remond and Pizette, 2014; Mechtcherine and Shyshko, 2015).

Also, the emergence of unprecedented technological advances in terms of automation and control processes is revolutionizing the construction industry. Of great importance is concrete 3D printing with its rapid development and a countless number of potential applications (Ghaffar et al., 2018; Wangler et al., 2019; Mechtcherine et al., 2019; Sanjayan and Behzad, 2019). For concrete 3D printing, the challenge is twofold; on the one hand, 3D printing technologies need to adapt to the rheological characteristics of concrete (Buswell et al., 2018). On the other hand, new materials for use with current 3D printing technologies need to be developed (Gosselin et al., 2016; Paul et al., 2017; Roussel, 2018). Moreover, the formulation and validation of analytical

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and computational approaches in this new branch of construction are crucial to deepen our understanding of both the printing process and the behavior of printed products.

This paper presents the Discrete Fresh Concrete Model (DFCM), which is a novel approach to simulate the behavior, i.e., transport, placement, and compaction, of fresh standard and printable concrete.

DISCRETE FRESH CONCRETE MODEL (DFCM)

Discrete (or Distinct) Element Method (DEM) was initially developed by Cundall and Strack (1979) for granular geomaterials. This approach, in its many variations, has since been adopted by many to simulate the behavior of many infrastructure materials such as concrete (Bolander et al., 1996; Cusatis et al., 2011a,b) and rock (Donzé et al., 2008; Scholtès and Donzé, 2013).

Following the original DEM approach, DFCM adopts rigid composite spherical particles of different sizes to simulate coarse stiff aggregate surrounding the layer of soft mortar (Figure 1). This approach is also known as the “excess paste theory", which was first introduced by Kennedy (1970) and then used widely by Puri and Uomoto (1998), Bentz et al. (1999), and Remond and Pizette (2014).

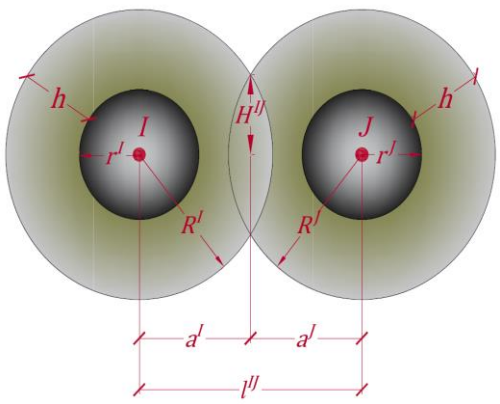

Figure 1. General configuration of two interacting composite particles.

A combination of parallel and series nonlinear springs and viscous dashpots describes the contact behavior of adjunct particles, and it considers all mechanical and rheological properties of fresh concrete. The constitutive equations that govern the spring and dashpot behavior include the normal and tangential components of the contact model. Each component takes into account (1) the viscous interaction between two interacting soft mortar layers and (2) the stiff interaction of two hard aggregate pieces.

Based on the stress-strain diagram shown in Figure 2a, the normal component of the targeted node-to-node contact model reflects both compression and tension modes in the interaction between the particles. It simulates the compressive behavior of aggregate to aggregate interactions and the compressive and tensile behavior of mortar to mortar layers interactions. Moreover, different sets of constitutive equations govern the tangential component between hard aggregates and soft excess paste. The mortar layer of fresh concrete was considered as a Bingham fluid with rheological characteristics (Figure 2b).

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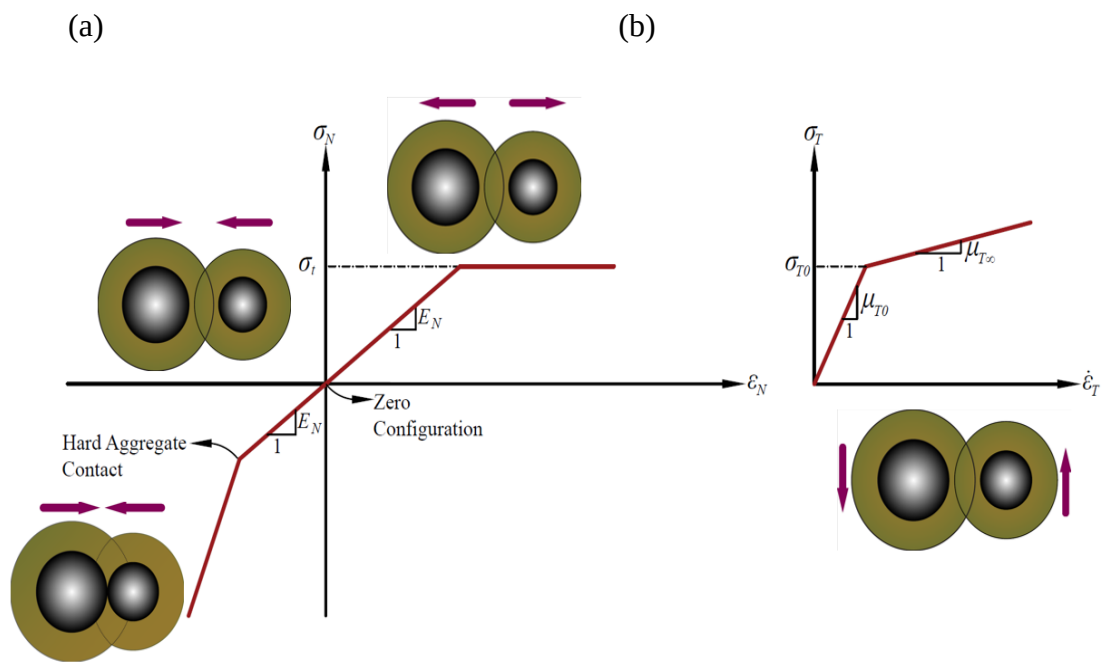

Figure 2. Diagrams of (a) Stress-strain for normal component, (b) Stress-strain rate for the tangential component of the mortar layer.

Unlike the conventional DEM with force-displacement based constitutive equations (e.g., Cundall and Strack, 1979), DFCM constitutive equations have been formulated in the form of stress-strain relationships. Such strategic turn allows for describing the fresh concrete behavior by those parameters also used for modeling hardened concrete.

Needless to say, the parameter identification for any newly developed models is a necessity for the successful reproduction of experimental results. Trial and error is the first choice in the calibration process, which is computationally intensive in the case of discrete element simulations. However, the methodology for the proper estimation of key parameters of DFCM was also investigated to facilitate the calibration process. The key parameters, based on stress-strain and strain rate diagrams shown in Figure 2, are the tensile strength (σ_t), the mortar normal modulus (E_N), the shear yield stress (σ_{T_0}), and the plastic viscosity (μ_∞). The mortar thickness (h) (Figure 1) is geometrically significant to calculate the normal strain (ε_N), and shear strain rate ($\dot{\varepsilon}_T$) (Figure 2). Based on a series of sensitivity analyses, the tensile strength can be estimated numerically from the lateral pressure of fresh concrete on the formwork. The mortar elastic modulus and the mortar thickness can be obtained from experimental data on volumetric fraction, density, and bulk modulus ($p = \Delta V / V_0$). The shear yield stress and plastic viscosity can be identified by experimental test measuring flowability, viscosity, and workability. Also, DFCM needs some other parameters such as the density of mortar and aggregate, their Poisson's ratio, and aggregate elastic modulus (the slope of the second branch of the normal stress-strain curve in compression mode of Figure 2a), that can be determined, typically, from the available literature or technical specifications.

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REPRODUCING EXPERIMENTAL RESULTS OF STANDARD AND PRINTABLE CONCRETE

With the insight obtained from the sensitivity analysis, two typical fresh concrete tests were chosen to validate the developed model: Slump and ICAR rheometer tests. As shown in Figure 3, DFCM succeeds in generating accurate results of high, medium flowable, and very stiff concrete. Figure 3b shows the result of a slump test simulation for self- consolidating concrete (SCC), which based on most of the experiments, can spread in a range of 45.5 to 81.0 cm after removing the slump cone. The simulated SCC (Figure 3b) with a 60 cm diameter and 25.0 cm slump, is an appropriate sample that represents the capability of DFCM to reproduce the flowability of SCC. Figure 3c shows a slump test simulation for a typical ordinary concrete with medium workability and a true slump in shape. Finally, according to Wolfs et al. (2018), printable concrete should have only minimal slump and that shown in Figure 3d in which the numerical simulation provides a decrease in height and horizontal spread of only a couple of centimeters.

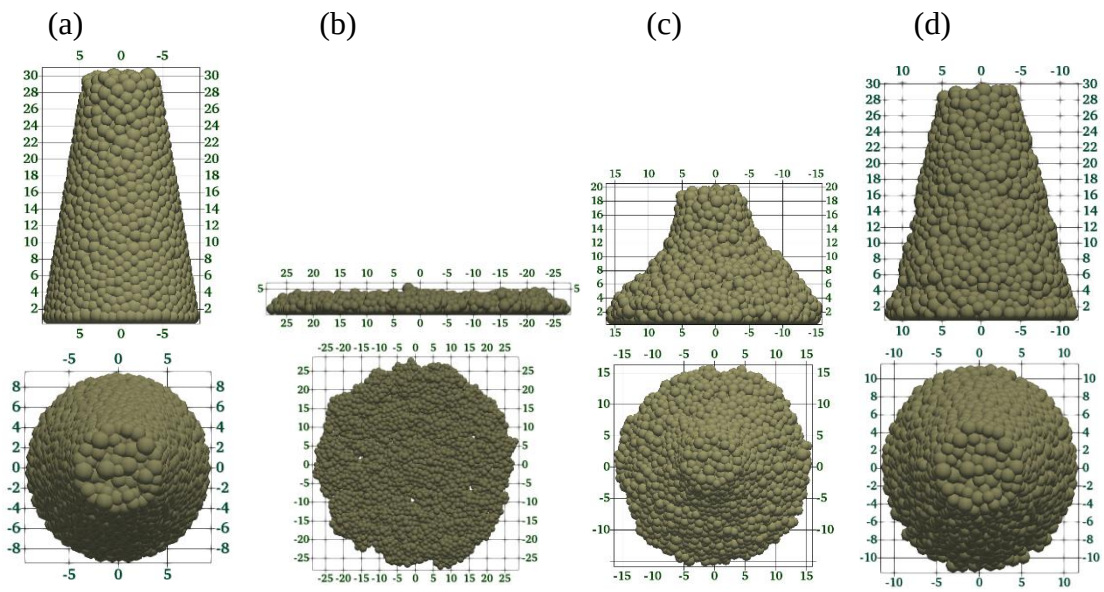

Figure 3. Side view and top view of (a) Prepared sample for the slump slow test; Result of the slump flow test for typical (b) SCC, (c) Ordinary cement concrete, (d) Printable concrete (unit of length is cm).

Similarly, as shown in Figure 4, DFCM reproduces well the experimental results of the ICAR rheometer test for the regular concrete (González-Taboada et al., 2017) and printable concrete (Al-Chaar et al., 2018). Figure 4a shows a slice of the simulated sample before the test starts; Both Figures 4b and 4c are the final configurations of typical ordinary and printable concrete, respectively, that are in agreement with relevant experimental observations. For regular concrete (Figure 4b), particles flow all around the container smoothly during the test. However, for printable concrete, Figure 4c shows that during the test solid-like pieces of material move around the vane and most particles do not move and actual "flow" cannot be identified.

The obtained vane resistant torque versus applied rotational speed for the regular and two different mix design of printable concrete is matched with experiments in Figures 4d and 4e.

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According to Al-Chaar et al. (2018), printable concrete at the fresh state is a less rate-dependent concrete than standard concrete as the entirely different response in the ICAR test shows in both experiments and simulations.

Therefore, in selecting the correct parameter sets for reproducing these experimental results, efforts were devoted to reducing the impact of rate-dependent parameters of DFCM on the ICAR test simulation results. Finally, the result shown in Figure 4e was obtained by selecting low values for the plastic viscosity and the viscous damping in both the tangential and normal components and high values for the shear yield stress.

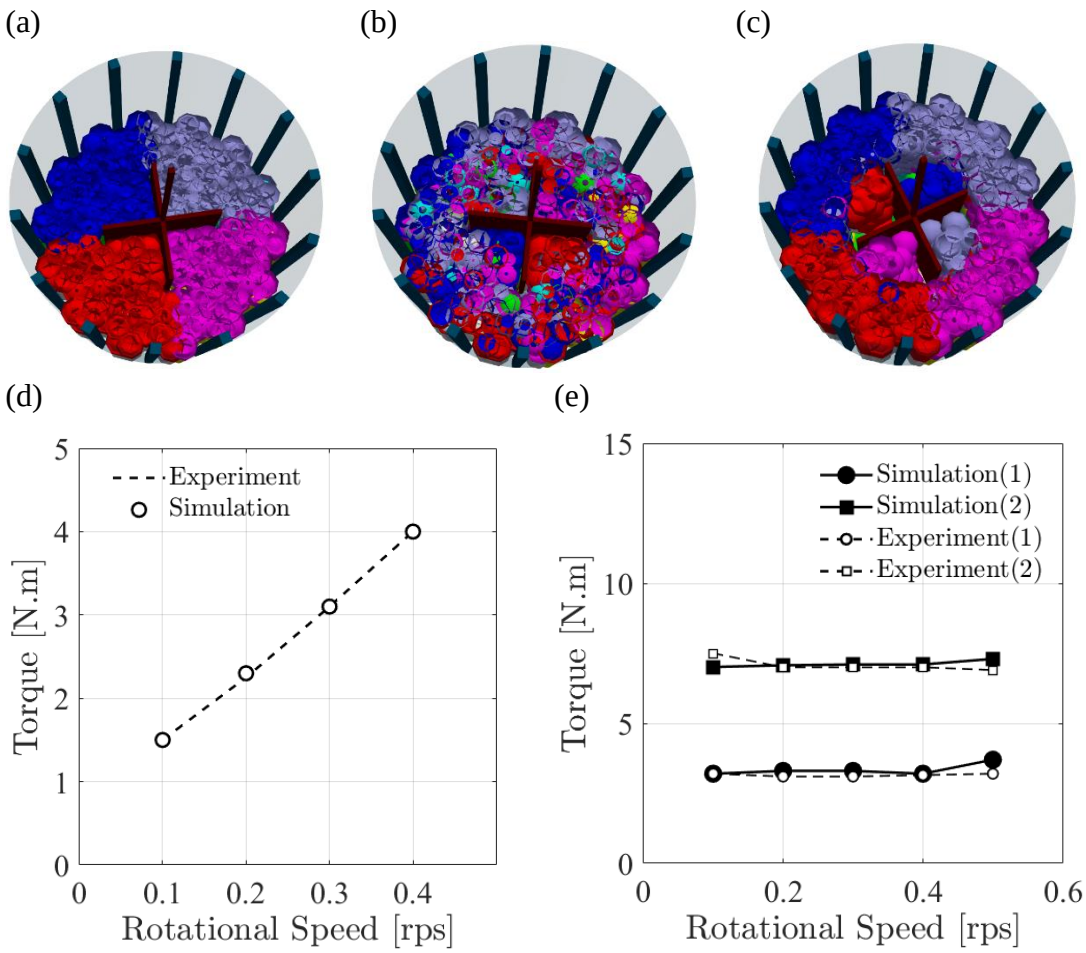

Figure 4. A configuration of (a) The prepared numerical sample before the test (b) The end of the test of typical standard concrete, (b) The printable concrete at the end of the test; The vane resistant torque vs applied rotational speed of (d) The typical standard concrete, (e) The printable concrete

NUMERICALLY 3D PRINTING OF THE NORTHWESTERN MARTIAN HABITAT

Northwestern University in collaboration with Skidmore, Owings & Merrill (SOM) designed a dome-shaped Martian habitat as a part of NASA's 3D-Printed Habitat Challenge (Figure 5a).

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The habitat has been designed specifically to be constructed by using 3D printing technologies. For the purpose of control, prediction, optimization of the printing process, material flow, and the final configuration of the printed objects, the 3D printing process can be monitored with a numerical model. The development of this numerical model is essential, especially for a long-term extraterrestrial landing in which any construction processes should be accomplished by autonomously remote operations, with no-human involvement.

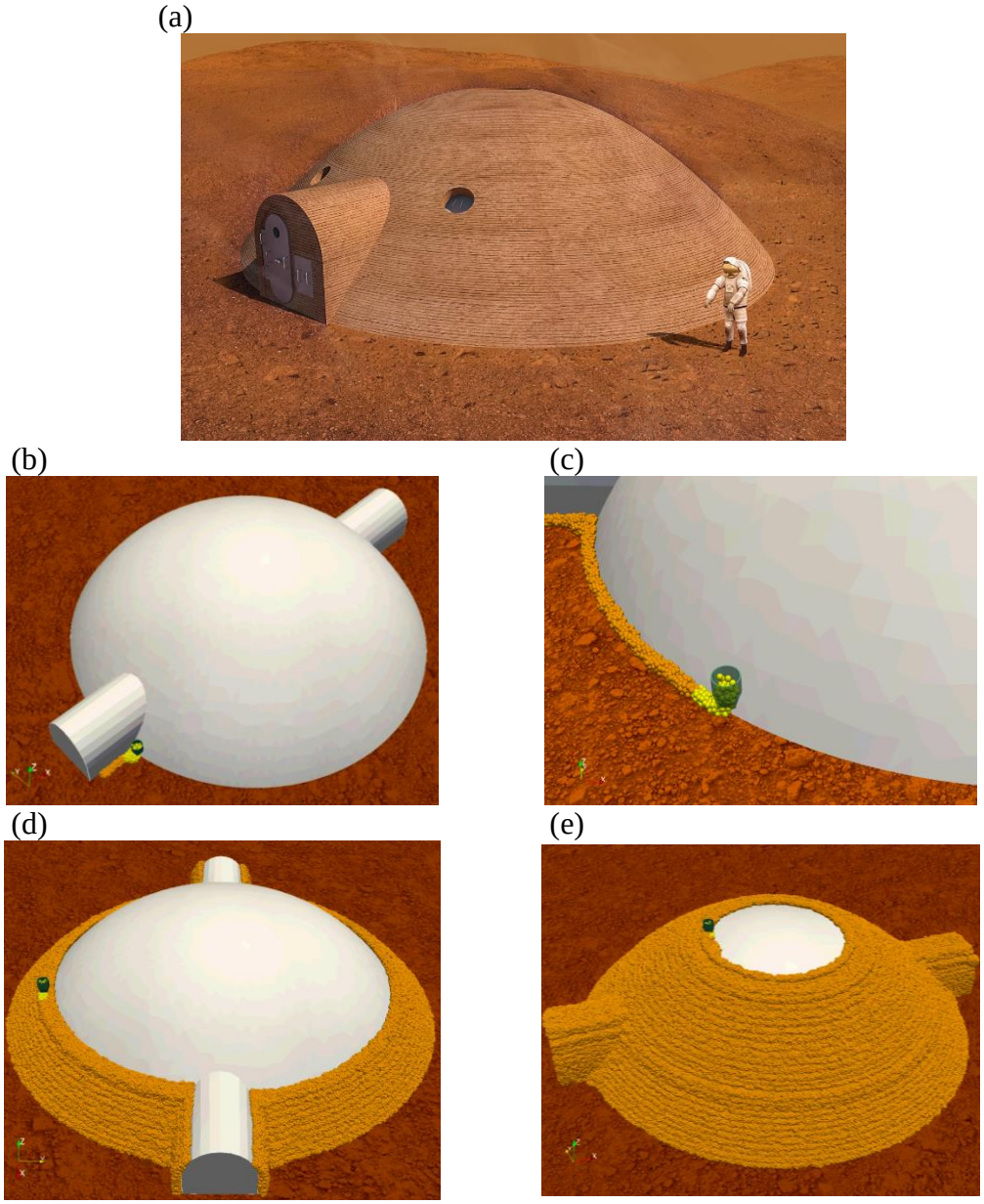

Figure 5. Northwestern Martian Habitat (a) Exterior view, (b) to (e) Numerically layer by layer printing

For the development the numerical model for printing, the motion of the printer component (the nozzle) was synchronized with the material generator (DEM particle generator) in which

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DFCM governs the microscopic rheological and tribological behavior of inter-particle and surface-particle (nozzle-particle) interactions, respectively. For simulation of the 3D printing process on planets, velocity of nozzle should be compatible with the discharge rate of the material. Obviously, discharge rate is dependent on the applied gravity and pump pressure for which inter-particle rheological characteristics (DFCM contact parameters) can be identified with a set of sensitivity analyses based on existing data from equivalent experiments. As shown in Figure 5b-e, DFCM was used to perform the simulation of the 3D printing process for the Northwestern Martian Habitat. In this figure various stages of the printing process have been illustrated.

CONCLUSION

This paper presents a novel discrete model for simulation of fresh concrete for 3D printing applications. The developed contact model can satisfactorily simulate the rheological behavior of cementitious materials right after pouring. The results of the simulations demonstrate that DFCM is an accurate tool for modeling the behavior of self-compacting concrete, ordinary concrete, as well as printable concrete. Indeed, the numerical simulations of ICAR and slump tests are in complete agreement with relevant experimental tests. Finally, DFCM can be used successfully to simulate the 3D printing process of concrete structures on any planets.

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